

Success-Based System Development Business

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Opening

Hello everyone. Nice to meet you. I am Naohiko Hirano of Hirano System Design Office. Thank you for giving me the opportunity to speak today. I would like to explain the concept of a success-based system development business: moving away from man-month billing and labor-intensive work toward knowledge-intensive work.

Business concept and background

The business is based not on conventional man-month billing, but on improvements in the customer's profits generated by the effects of system investment. Japan's international competitiveness had been declining relatively, and profitability in the IT industry was lower than in Europe, the United States, and India. In addition, an aging and declining population and the retirement of the baby-boomer generation could make it difficult to maintain large systems built through labor-intensive approaches. The rapid transition toward a digital society also makes it necessary for organizations to consider the balance of their IT utilization.

Contract-based development continues to rely on a multilayer subcontracting structure and man-month billing. In particular, subcontracted engineers can become an adjustment mechanism for economic fluctuations and may face worsening working environments, employment conditions, and compensation. These people nevertheless support Japan's IT infrastructure.

Why the business is needed

From the customer's perspective, the objective is to increase organizational value quickly while determining the appropriate balance of IT investment. From the development side, the objective is to move from labor-intensive to knowledge-intensive work, create mechanisms for individual and organizational learning, and transform the organization so that it can generate revenue and innovation without depending on man-month billing. From the perspective of the industry as a whole, this can contribute to the development of industries in Japan and overseas.

Why a success-based model can address the issue

Under man-month billing, revenue is proportional to the number of employees. When orders decline, the business can face a cash-flow shortage, and unlike manufacturing, recovery through the next sale is not readily possible. Unit prices per man-month were also declining. Under a success-based model, customer profit is directly related to revenue, so the number of employees and revenue do not have to be proportional. People still perform the work, but knowledge and judgment are essential. To move beyond man-month billing and labor-intensive work, the revenue structure itself must change.

Business process

In the conventional model, customer requirements are converted into specifications, an estimate is prepared using man-months, a contract is concluded, development and testing are performed, and the project is considered complete after operation is confirmed. After that, interest in the customer's ongoing results may diminish, and a multilayer subcontracting structure can prevent both customer knowledge and development capabilities from increasing at the organizational level.

In a success-based model, the customer's post-operation profit becomes an important concern. The process therefore includes profit-contribution planning, IT-contribution balance analysis, effect-verification planning, investment recovery, gain-sharing planning, and risk planning. The perspective must change from simply building a

system to achieving outcomes.

Contract concept and game theory

The success-based contract applies game theory. Situations in which benefits are exchanged within mutual interdependence are treated as games. A game in which players independently choose strategies is a non-cooperative game, while a game in which players can cooperate through alliances is a cooperative game. The success-based approach is based on the cooperative-game concept. In cooperative games, an inductive approach is used to identify similarities among observations under an assumed desirable outcome and derive conclusions.

Three principles for negotiation and verification

How should negotiations be conducted to make the maximum gain available to the parties? There are three main principles. First, clarify what had been a black box within the company, especially the roles and target figures relating to IT and non-IT activities. Second, eliminate information asymmetry and bias. Third, clarify the boundaries of the enterprise. These principles are the key to moving from man-month billing and labor-intensive work toward knowledge-intensive work.

Performance evaluation

Evaluation based on the contract concept uses key performance indicators (KPIs) provided by strategy maps and the Balanced Scorecard to measure performance quantitatively. To evaluate trade-offs objectively, methods such as Activity-Based Costing (ABC), Resource Consumption Accounting (RCA), and Customer Relationship Management (CRM) information are used.

People, IT, equipment, and other assets are resources. The activities performed by these resources form the basis for evaluation, reporting, analysis, and planning. Activities are central to Activity-Based Management (ABM), which can be used to measure expenses and customer profitability accurately. These concepts can also be applied to organizations with purposes other than profit, including public institutions, hospitals, and schools, by focusing on the relevance and appropriateness of all activities.

Resources supply costs to activities, while cost objects consume activities. Costs combine manufacturing, provision, and service costs. These figures become inputs to KPIs and are used to communicate strategy and determine whether the organization is achieving its expected results.

Integrated analytical system

Implementation requires a system for analyzing organizational performance that integrates methodologies and their integration, ontology, data science, inference, and expert systems. By obtaining new insights through this evaluation, the organization can move toward knowledge-intensive work.

High-performance organization

A high-performance organization is necessary to support this transformation. Its objectives include high added value, high quality, short development cycles, low cost, and, for some systems, improved safety. People are the foundation. Individuals need to develop specialized knowledge, use tools effectively, increase reuse, and re-engineer existing systems. The evaluation of technologies and improvement of measurement methods are also necessary.

Man-month billing reflects a Taylorist approach: managers define tasks and workers are expected to complete them as quickly as possible. Traditional measures such as lines of code per day and working hours have therefore been used. The question is whether such measures can lead an organization toward high performance.

People, organizations, and innovation

The key to becoming knowledge-intensive lies in people and organizations. It is often assumed that people make an organization intelligent, but the reverse can also be true: organizations create systems that shape and amplify how people work. Our lives are enriched by individual talent, but organizations operate under different rules. They

execute and coordinate the efforts of many people. Organizations that succeed at this have highly effective systems.

The speed at which complex problems can be understood and solved is determined not only by skills and abilities but also by the environment: organizational openness, whether completion is valued over outcomes, and whether people have enough room to try new ideas and technologies and to talk with one another.

Motivation and organizational culture

In labor-intensive work, extrinsic motivation such as bonuses may be effective. In knowledge-intensive work, intrinsic motivation may play a greater role. Autonomy, mastery, and purpose are often identified as important factors. One of the most important responsibilities of a leader is building organizational culture.

Culture can be highly stable and difficult to change. It represents what a group has learned and accumulated about how to think, feel, and perceive the world in pursuing success. To change culture, an organization must decide what it wants to achieve, how it wants people to behave, and what people want from one another; provide training; and reinforce desired behavior. As culture changes, individuals can grow and constructive dialogue can emerge, making innovation easier to create.

Closing

The Bible contains the expression, "New wine cannot be put into old wineskins." Perhaps the same can be said of organizational systems.

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